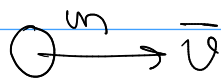


Friday 9 P.M.

Linear Momentum and collision

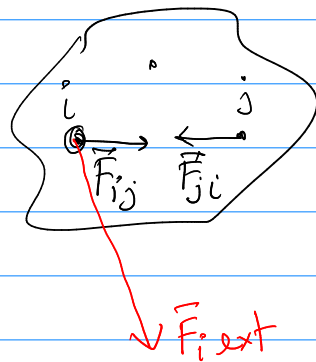


$$\boxed{\vec{p} = m\vec{v}} \quad \text{kg-m/s}$$

$$\Rightarrow \frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v})$$

$$= \frac{d\vec{p}}{dt} = m \frac{d\vec{v}}{dt} = m \vec{a} = \vec{F}_{\text{net}}$$

$$\Rightarrow \boxed{\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt} = \frac{\Delta \vec{p}}{\Delta t}}$$

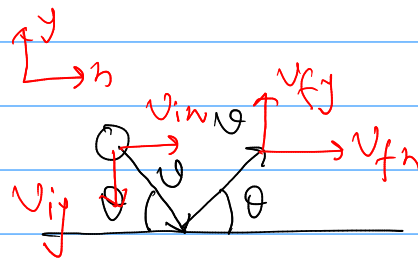


$$\text{If } \vec{F}_{\text{net}} = 0$$

$$\Delta \vec{p} = 0$$

$$\vec{p}_f - \vec{p}_i = 0$$

$$\Rightarrow \boxed{\vec{p}_f = \vec{p}_i}$$



$$\Delta \vec{p} = ?$$

$$|v_{ix}| = |v_{fx}|$$

$$|v_{iy}| = |v_{fy}|$$

$$\Delta \vec{p}_x = mv_{fx} - mv_{ix}$$

$$\boxed{\Delta p_x = 0}$$

$$\Delta \vec{p}_y = mv_{fy} - m(-v_{iy})$$

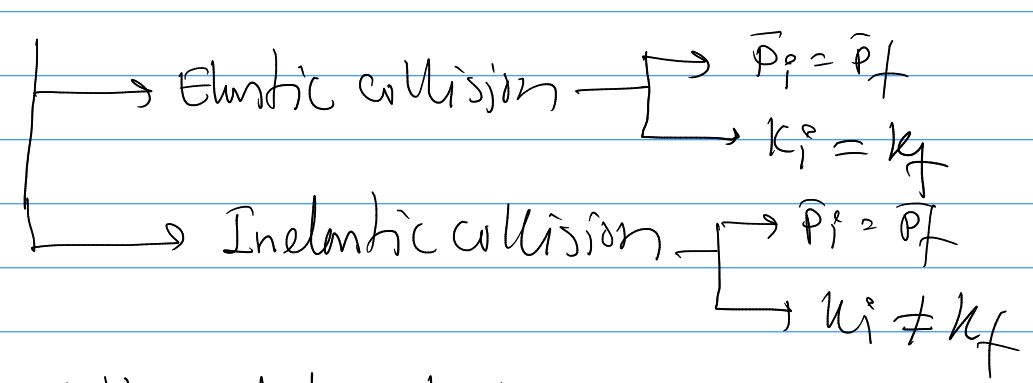
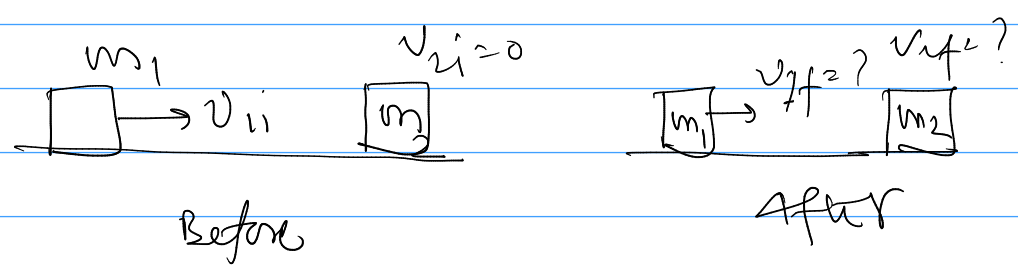
$$\Delta p_y = mv_{fy} + mv_{iy}$$

$$\Delta \vec{p}_y = 2mv_{iy} \neq 0$$

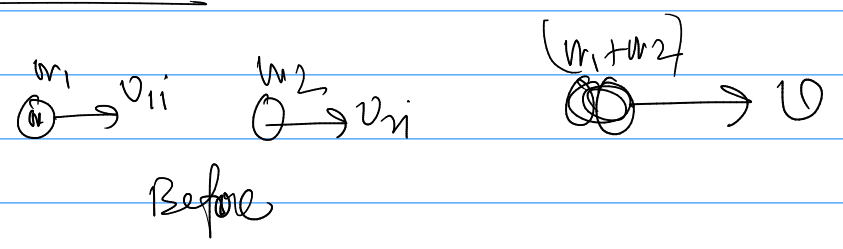
$$\boxed{\Delta \vec{p} = p_x \hat{i} + p_y \hat{j}}$$

$$\boxed{\bar{p}_{i1} + \bar{p}_{i2} + \dots + \bar{p}_{in} = \bar{p}_{f1} + \bar{p}_{f2} + \dots + \bar{p}_{fn}}$$

Collision:



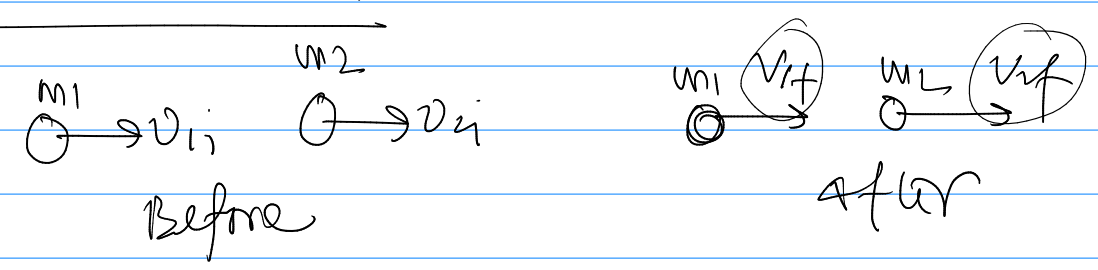
Completely inelastic collision:



$$\boxed{\bar{p}_i = \bar{p}_f}$$

$$\Rightarrow \boxed{m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v}$$

Collision in 1D and 2D:-



$\bar{p}_i = \bar{p}_f$

$$\Rightarrow \boxed{m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}} \quad \text{--- (1)}$$

$$\# \quad K_i = K_f$$

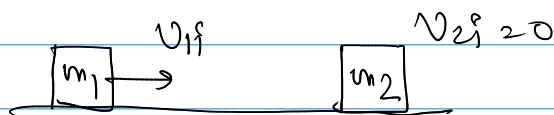
$$= \left[\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 \right] \quad (11)$$

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i} + \frac{2m_2}{m_1 + m_2} v_{2i}$$

$$v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i} + \frac{m_2 - m_1}{m_1 + m_2} v_{2i}$$

Case-5:

$$m_1 > m_2 \quad v_{2i} = 0$$



$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i} + \frac{2m_2}{m_1 + m_2} v_{2i}$$

$$v_{1f} = 0$$

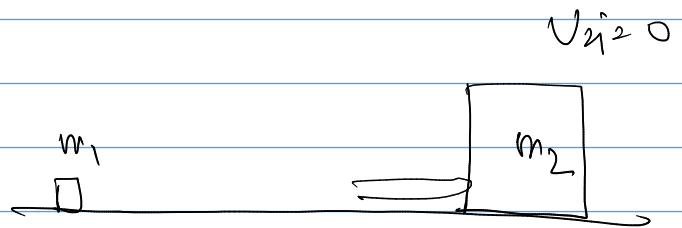
$$v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i} + \frac{m_2 - m_1}{m_1 + m_2} v_{2i}$$

$$v_{2f} = \frac{2m_1}{2m_1} v_{1i}$$

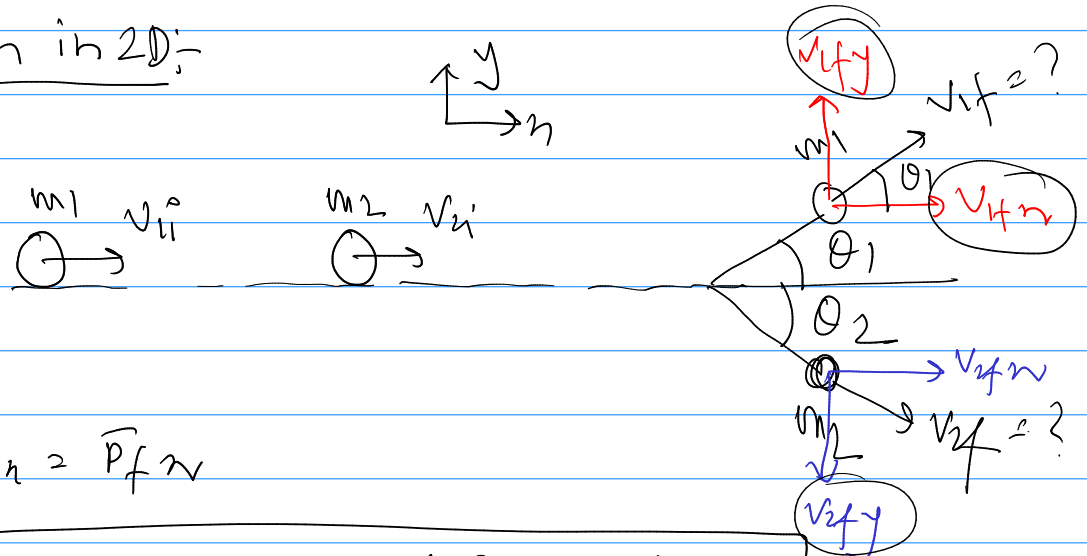
$$v_{2f} = v_{1i}$$

$m_1 \gg m_2$ $v_{2i} = 0$

$m_2 \gg m_1$



Collision in 2D:-



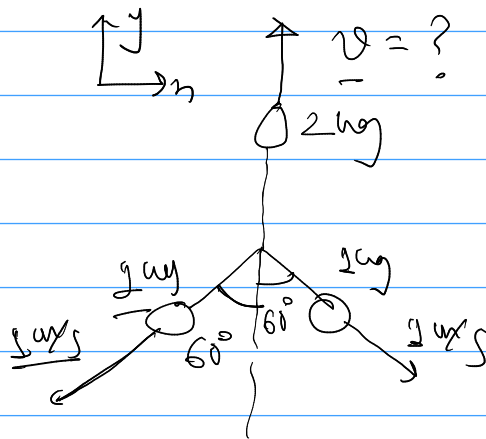
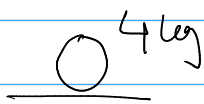
$\vec{P}_{ix} = \vec{P}_{fx}$

$\Rightarrow m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} \cos \theta_1 + m_2 v_{2f} \cos \theta_2$ — (1)

$\vec{P}_{iy} = \vec{P}_{fy}$

$\Rightarrow 0 = m_1 v_{1f} \sin \theta_1 - m_2 v_{2f} \sin \theta_2$ — (2)

THUR - 9:00 PM



$$P_{iy} = P_{fy}$$

mv

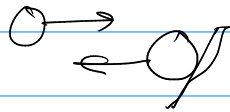
$$\Rightarrow 0 = P_{fy}$$

$$\Rightarrow 0 = 2 \times v - 1 \times 1 \times \cos 60^\circ - 1 \times 1 \times \cos 60^\circ$$

$$\Rightarrow \boxed{v = 0.5 \text{ m/s}}$$

$$\boxed{\vec{v} = 0.5 \hat{j} \text{ m/s}}$$

Impulse:

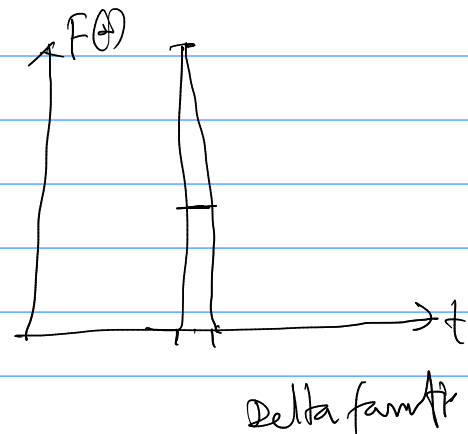


$$\vec{F}_{int} = \frac{\Delta \vec{p}}{\Delta t}$$

$$\Rightarrow \Delta \vec{p} = \vec{F}_{int} \times \Delta t$$

$$\Rightarrow \vec{p}_f - \vec{p}_i = \vec{F}_{int} \times \Delta t$$

$$\Rightarrow \vec{I} = \vec{F}_{int} \times \Delta t$$

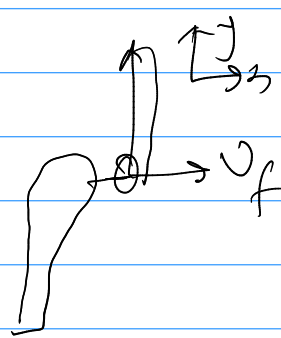


$$\boxed{\vec{I} = \Delta \vec{p} = \vec{p}_f - \vec{p}_i = \vec{F}_{int} \times \Delta t}$$

$$\checkmark I = \Delta \vec{p} = \vec{p}_f - \vec{p}_i = \vec{F}_{av} \times \Delta t$$

$$\checkmark I = \int_{t_1}^{t_2} \vec{F} dt$$

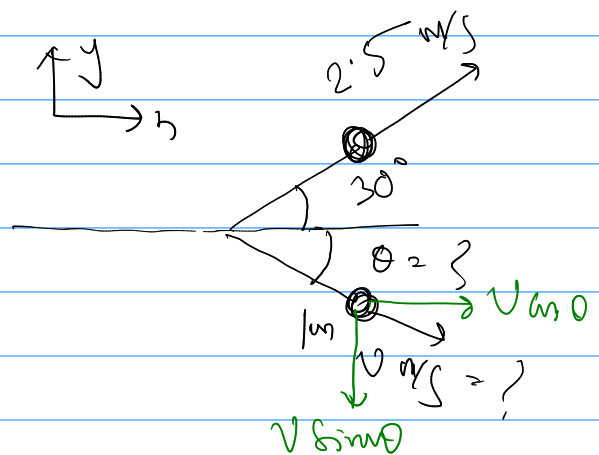
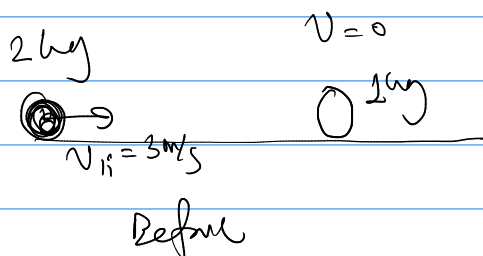
$v = 60 \text{ m/s}$, $F_{av} = ?$ $\vec{F}_{av} = \frac{\Delta \vec{p}}{\Delta t} = \frac{\vec{p}_f - \vec{p}_i}{\Delta t}$
 $m = 4 \text{ gm}$
 $\Delta t = 2 \text{ ms}$



$$F_{av} = \frac{mv_f - mv_i}{\Delta t}$$

$$F_{av} = \frac{4 \times 10^{-3} \times 60}{2 \times 10^{-3}}$$

$$F_{av} = 120 \text{ N}$$



$$\# \vec{p}_{ix} = \vec{p}_{fx}$$

$$2 \times 3 + 1 \times 0 = 2 \times 2.5 \cos 30^\circ + 1 \times v \cos \theta$$

$$v \cos \theta = 1.67 \text{ m/s} \quad \text{--- (i)}$$

$$\# \vec{p}_{iy} = \vec{p}_{fy}$$

$$0 = 2 \times 2.5 \sin 30^\circ - 1 \times v \sin \theta$$

$$v \sin \theta = 2.5 \quad \text{--- (ii)}$$

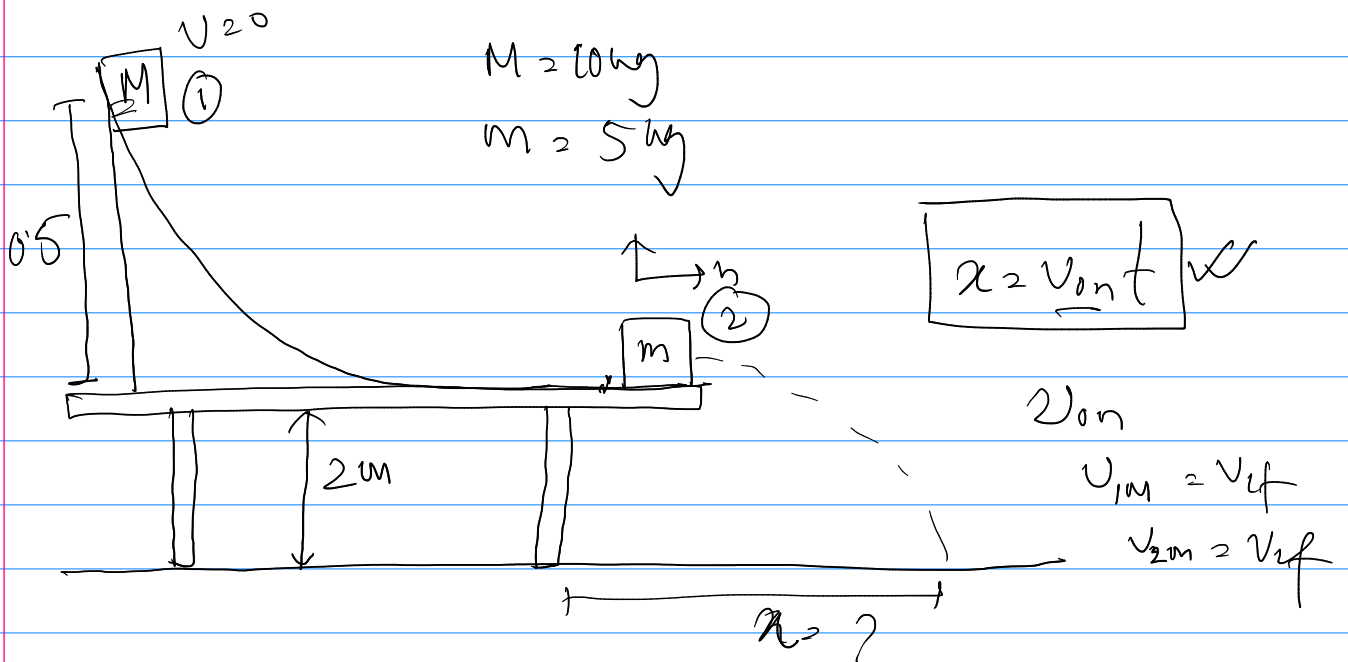
$$\sqrt{v_1^2 + v_2^2} = v = (1.57)^2 + (2.5)^2$$

$$= \boxed{v = 3.006 \text{ m/s}} \quad \underline{Ans}$$

$$\theta = \tan^{-1} \frac{2.5}{1.57} = 56.28^\circ \quad \underline{Ans}$$

$$\Delta t = 2 \text{ ms}$$

$$F_{av} = \frac{\Delta p}{\Delta t}$$



$$x = v_{0nt} \quad \checkmark$$

Don

$$v_{1m} = v_{1f}$$

$$v_{2m} = v_{2f}$$

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i} + \frac{2m_2}{m_1 + m_2} v_{2i}$$

$$m_1 = 10 \text{ kg}$$

$$m_2 = 5 \text{ kg}$$

$$v_{1f} = \frac{10 - 5}{10 + 5} 3.13 = 1.04 \text{ m/s}$$

$$v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i} + \frac{m_2 - m_1}{m_1 + m_2} v_{2i}$$

$$v_{1i} =$$

$$v_{2i} = 0$$

$$E_i = E_f$$

$$K_i + U_i = K_f + U_f$$

$$mgh = \frac{1}{2}mv_i^2$$

$$v_i = \sqrt{2gh} = 3.1 \text{ m/s}$$

$$v_{yf} = \frac{2 \text{ m}}{m_1 + m_2} v_{yi} = \frac{2 \times 10}{15} 3.13 \checkmark$$

$$\boxed{v_{yf} = 4.17 \text{ m/s}}$$

$$y = y_0 + v_{oy}t + \frac{1}{2}a_yt^2 \checkmark$$

$$-2 = 0 + 0 - 4.9t^2 \checkmark$$

$$\Rightarrow \boxed{t = \sqrt{\frac{2}{4.9}} = 0.63 \text{ s}}$$

$$\boxed{x_M = v_{ox}t = 1.04 \times 0.63 = 0.65 \text{ m}}$$

$$\boxed{x_m = v_{mx}t = 4.17 \times 0.63 = 2.627 \text{ m}}$$